

AL MAKTOUM INTL, United Arab Emirates

WMO: 412000

Lat: 24.897N

Lon: 55.161E

Elev: 62

StdP: 14.66

Time Zone: 4.00 (E04)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.9	53.9	30.0	24.2	87.1	33.8	28.4	83.5	24.3	69.4	22.3	71.4	6.6	110	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.3	113.2	73.0	111.2	73.1	109.2	73.5	85.8	95.1	84.6	94.2	83.5	93.7	10.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.0	178.5	91.3	82.4	169.3	91.2	80.8	160.2	90.8	50.5	94.7	49.1	93.9	47.7	94.2	90.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.0	20.5	18.5	DB	47.0	117.7	2.2	1.1	45.4	118.5	44.1	119.1	42.8	119.7	41.2	120.5	
			WB	42.0	88.0	2.2	1.0	40.4	88.8	39.2	89.4	38.0	89.9	36.4	90.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.1	66.4	68.4	74.0	82.5	90.1	94.1	98.4	97.9	93.1	86.2	76.3	69.2
	DBStd	12.03	3.34	5.16	5.70	5.22	4.09	3.54	3.12	2.68	2.82	3.03	3.90	3.43
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	48	23	17	3	0	0	0	0	0	0	0	0	5
	CDD50	12090	508	516	744	974	1242	1322	1499	1485	1294	1122	790	594
	CDD65	6662	65	113	282	524	777	872	1034	1020	844	657	340	134
	CDH74	94786	468	867	2544	6159	11426	13672	17233	16877	12930	8560	3160	890
	CDH80	61745	68	284	1111	3368	7367	9416	12769	12416	8701	4853	1249	143
Wind	WSAvg	9.0	8.3	9.5	9.9	9.6	10.0	9.9	9.7	9.4	8.5	8.0	8.0	7.7
Precipitation	PrecAvg	3.4	0.6	0.6	0.8	0.3	0.1	0.0	0.2	0.1	0.1	0.1	0.1	0.4
	PrecMax	8.0	2.7	3.5	3.6	1.9	0.2	0.4	2.2	0.3	0.6	0.6	1.1	2.3
	PrecMin	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.7	0.8	1.0	0.4	0.0	0.1	0.4	0.1	0.1	0.1	0.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.5	92.8	98.7	104.4	111.0	113.4	116.5	115.1	111.6	103.9	95.2	86.3
		MCWB	63.1	63.2	66.3	67.3	69.1	72.1	73.1	72.3	71.7	69.7	66.5	64.4
	2%	DB	82.2	87.4	94.0	101.9	107.7	111.0	113.3	113.0	109.3	100.8	91.8	83.8
		MCWB	62.4	62.9	64.3	66.3	70.3	72.3	73.5	73.1	71.7	70.1	67.6	65.0
	5%	DB	79.0	83.2	89.9	98.7	105.5	107.9	111.2	110.9	106.2	98.8	89.6	81.0
		MCWB	61.5	62.1	63.4	66.2	69.3	73.3	74.3	73.4	72.4	71.1	67.6	65.3
	10%	DB	76.4	79.1	86.2	95.3	102.5	105.5	108.5	107.9	104.0	96.6	87.4	79.0
		MCWB	62.1	61.8	63.7	66.3	70.0	74.2	75.4	75.0	74.0	72.0	68.0	64.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	68.6	72.0	75.9	81.0	85.2	87.2	87.7	85.4	83.0	75.2	72.0
		MCDB	75.8	77.5	80.6	85.1	90.2	92.6	96.8	95.8	95.0	89.1	85.2	79.0
	2%	WB	66.8	67.2	69.9	73.9	79.4	83.6	85.7	86.0	83.9	81.0	73.5	70.1
		MCDB	75.0	76.2	79.5	83.7	89.1	92.5	95.5	95.1	92.8	89.1	83.3	77.1
	5%	WB	65.1	65.9	68.7	72.2	77.7	82.2	84.5	84.8	82.6	79.7	72.1	68.9
		MCDB	74.3	75.0	78.5	83.5	89.1	92.6	94.4	94.6	91.9	88.5	82.6	76.3
	10%	WB	63.7	64.5	67.4	70.4	75.9	80.8	83.4	83.3	81.5	78.4	70.6	67.1
		MCDB	72.7	74.3	77.6	83.7	89.3	92.4	94.5	94.5	91.2	87.6	81.4	75.2
Mean Daily Temperature Range	5% DB	MDBR	20.1	20.6	22.1	23.9	25.7	25.4	23.3	23.8	25.0	23.4	21.5	21.0
		MCDBR	24.8	25.6	27.5	28.7	29.2	28.1	25.9	26.5	28.0	25.6	24.3	23.1
		MCWBR	11.0	10.6	10.6	10.0	11.4	11.8	11.8	11.9	12.4	11.3	9.4	10.8
	5% WB	MCDBR	20.7	20.4	21.6	23.7	26.1	25.4	22.5	22.1	24.0	23.0	21.4	19.6
	MCWBR	11.2	10.2	10.3	10.4	11.1	11.0	10.7	11.3	11.2	10.3	9.3	9.9	
Clear-Sky Solar Irradiance	taub	0.445	0.503	0.566	0.598	0.589	0.639	0.730	0.698	0.601	0.511	0.456	0.437	
	taud	2.040	1.880	1.734	1.695	1.708	1.649	1.533	1.589	1.733	1.905	2.046	2.070	
	Ebn at Noon	251	245	237	233	234	222	202	208	225	239	246	248	
	Edh at Noon	48	60	73	78	77	81	90	85	72	58	47	45	
All-Sky Solar Radiation	RadAvg	1318	1566	1848	2113	2361	2403	2246	2161	2012	1756	1420	1246	
	RadStd	80	119	123	127	56	49	69	48	34	27	39	78	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	+0.85	+1.26	N/A	N/A	N/A	N/A	N/A	N/A	+270	+268

Nomenclature: See separate page